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Department of Computer Science

Faculty of Science and Technology

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Project Report On

Bakery Shop Management System

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Bakery Shop Management System

1. Project Description

In many small and medium-sized bakeries, daily operations such as managing perishable inventory, handling customer orders, and recording transactions are still conducted manually. This traditional approach often leads to human errors—such as selling expired items—delayed billing, data loss, and significant inefficiency in monitoring the stock levels of raw materials. The absence of an integrated system makes it difficult for owners to track staff performance, product availability, and total revenue in real-time. To address these challenges, the Bakery Shop Management System is proposed as a comprehensive desktop application developed in C#. This system aims to automate the core functions of bakery operations through a secure and user-friendly interface. It replaces manual system with a robust, database-driven solution that streamlines the entire workflow from production to the final point of sale. The system supports three primary roles—Admin, Inventory Manager, and Sales Cashier—each equipped with dedicated interfaces for managing staff, products, and sales transactions. By maintaining a centralized database, the application ensures data accuracy, prevents redundancy, and provides systematic control over all records. Furthermore, it ensures high levels of data integrity through rigorous input validation and verification protocols. Ultimately, this solution enhances operational efficiency, reduces manual workload, and provides real-time visibility into business performance, allowing for more informed decision-making.

2. Feature List

User_1 (Admin)

1. Login: Secure authentication for system access.
2. Logout: Securely exit the session.
3. Registration: Ability to register new user accounts for Admin, Managers and Cashiers.
4. Show Password: Toggle visibility for secure password entry.
5. Forget Password: A recovery mechanism that verifies the username against the database and retrieves the existing password if a match is found.
6. View Admin Information: Access and view the logged-in Admin's personal profile details.
7. Insert Employee: Add new staff records into the database.
8. Update Employee: Modify staff information.
9. Delete Employee: Remove inactive or resigned staff.
10. Search Employee: Find staff records.
11. Manage Category: Add, update, or delete product categories (e.g., Cakes, Breads).
12. View Sales Report: Real-time summary of total sales.
13. Account Status Control: Enable/Disable user access (Deactivate account) for security purposes.

User_2 (Inventory Manager)

1. Log In: Secure authentication for the Manager to log into the system using their username and password.
2. View Personal Profile: Allows the logged-in Manager to view and check their own profile details.
3. Employee Monitoring: Permits the Manager to view and monitor cashier employee records.
4. Add New Products: Allows inserting new bakery items into the database with details like price and stock levels.
5. Update Product: Allows modifying existing product data, including updating prices, names, or item descriptions.
6. Delete Products: Provides the option to remove items from the active product list if they are discontinued or out of stock.
7. Product Search System: A quick search feature to find specific items in the inventory easily using names or filters.
8. Inventory & Stock Control: Tracks and updates available stock levels in real-time to maintain a proper supply of ingredients or items.
9. View Sales Reports: Allows the Manager to view sales summaries and total revenue generated to check business performance.
10. Log out: Securely exit the system and terminate the active session.

User_3 (Sales Cashier)

1. Secure login and logout system: Access the sales terminal securely and close the session post-shift.
2. View Personal Profile: Access and view the logged-in Cashier's profile details.
3. Add Products to Cart: Select and add multiple bakery items to a virtual shopping cart based on the customer's order.
4. Modify Cart: Adjust item quantities or remove products from the cart before final processing.
5. Process Orders: Confirm and finalize the transaction once the customer completes the purchase.
6. Automatic Bill Calculation: Automatically compute the total amount, including item rates, subtotals, and applicable taxes or discounts, eliminating manual calculation errors.

3. Entity Relationship Diagram

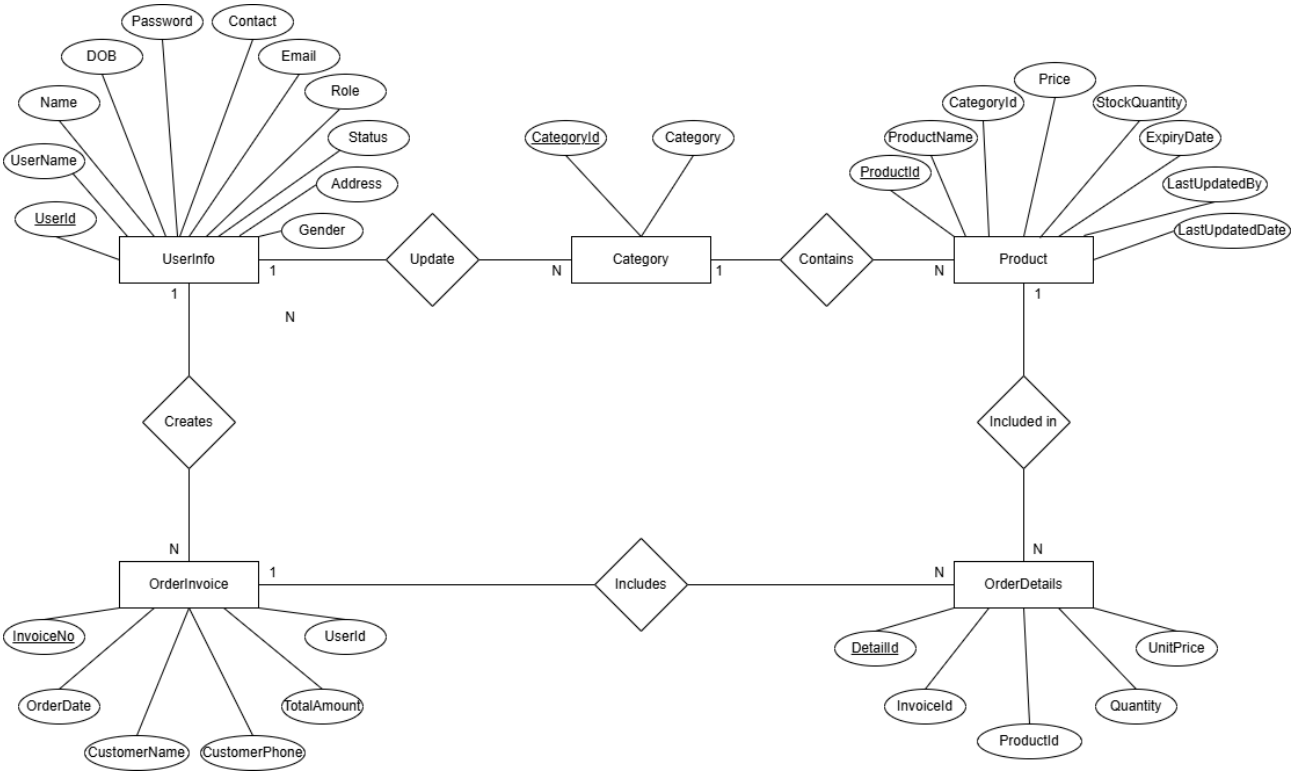


Fig-1: ER Diagram representing all entities, attributes, and relationships in the system

Normalization

1.UserInfo Table

- 1NF: UserId (PK), Name, UserName, Password, DOB, Contact, Email, Role, Status, Gender, Address.
- 2NF: UserId (PK), Name, UserName, Password, DOB, Contact, Email, Role, Status, Gender, Address.
- 3NF: UserId (PK), Name, UserName, Password, DOB, Contact, Email, Role, Status, Gender, Address.

2.Category Table

- 1NF: CategoryId (PK), CategoryName
- 2NF: CategoryId (PK), CategoryName
- 3NF: CategoryId (PK), CategoryName

3.Product Table

- 1NF: ProductId (PK), ProductName, CategoryId, Price, StockQuantity, ExpiryDate, LastUpdatedBy, LastUpdatedDate
- 2NF: ProductId (PK), ProductName, CategoryId (FK-Category.CategoryId), Price, StockQuantity, ExpiryDate, LastUpdatedBy (FK-UserInfo.UserId), LastUpdatedDate
- 3NF: Same as 2NF.

4.OrderInovoice Table

- 1NF: InvoiceNo (PK), OrderDate, CustomerName, CustomerPhone, TotalAmount, UserId
- 2NF: InvoiceNo (PK), OrderDate, CustomerName, CustomerPhone, UserId (FK-UserInfo.UserId), TotalAmount
- 3NF: Same as 2NF

5.OrderDetails Table

- 1NF: DetailId (PK), InvoiceNo, ProductId, Quantity, UnitPrice
- 2NF: DetailId (PK), InvoiceNo (FK-OrderInvoice.InvoiceNo), ProductId (FK-Product.ProductId), Quantity, UnitPrice
- 3NF: Same as 2NF.

Finalization:

- UserInfo: UserId (PK), Name, UserName, Password, DOB, Contact, Email, Role, Status, Gender, Address
- Category: CategoryId (PK), CategoryName
- Product: ProductId (PK), ProductName, CategoryId (FK), Price, StockQuantity, ExpiryDate, LastUpdatedBy (FK), LastUpdatedDate
- OrderInvoice: InvoiceNo (PK), OrderDate, CustomerName, CustomerPhone, TotalAmount, UserId (FK)
- OrderDetails: DetailId (PK), InvoiceNo (FK), ProductId (FK), Quantity, UnitPrice

4. Form Designs Screenshots

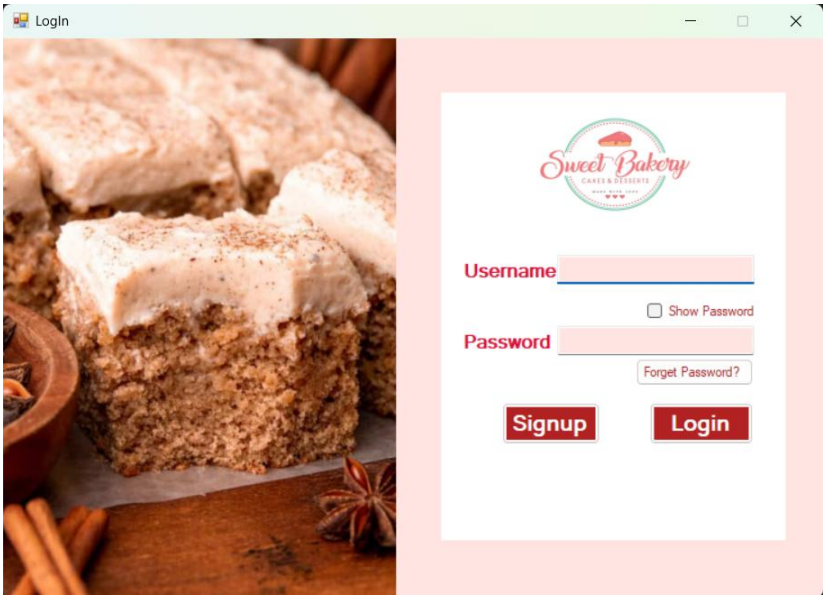


Fig-2: Login Form - Allows Admin, Manager and Cashier to sign in securely, with password visibility.

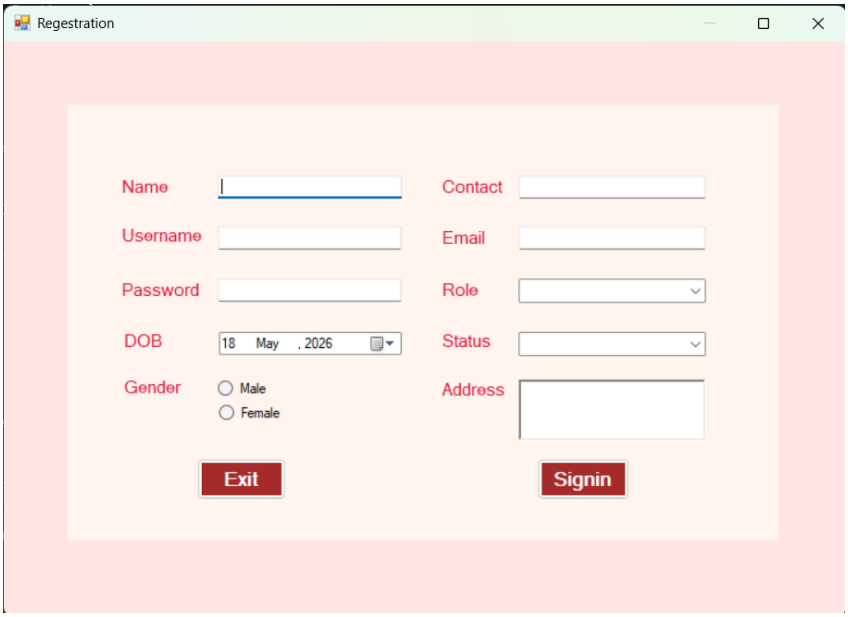


Fig-3: Registration Form - Allows Admin, Manager and Cashier to register new account.

Admin Accessed Form

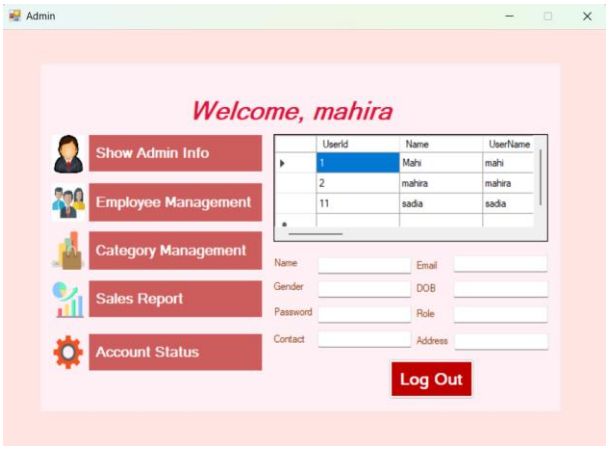


Fig-4: Admin Dashboard

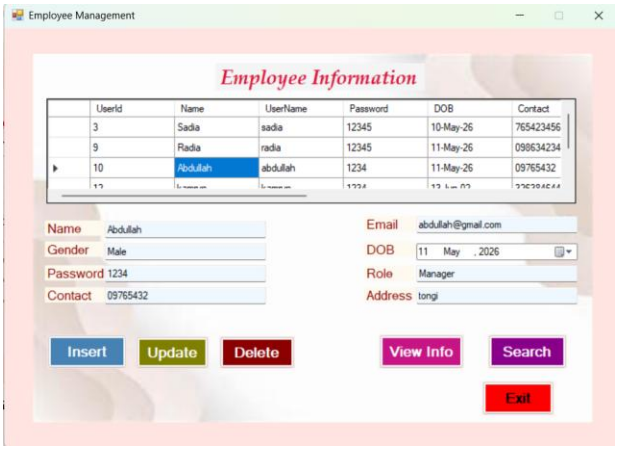


Fig-5: Employee Management

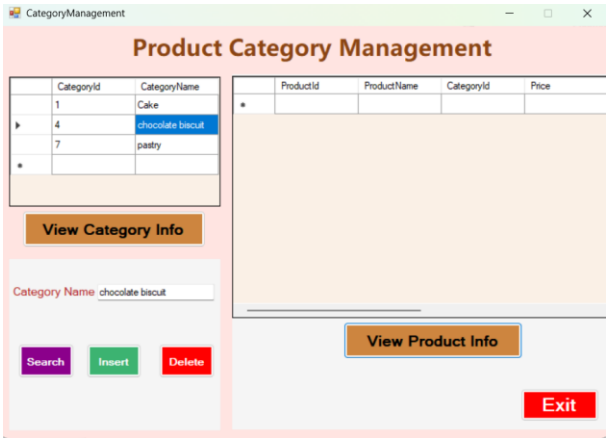


Fig-6: Admin Product Category Management

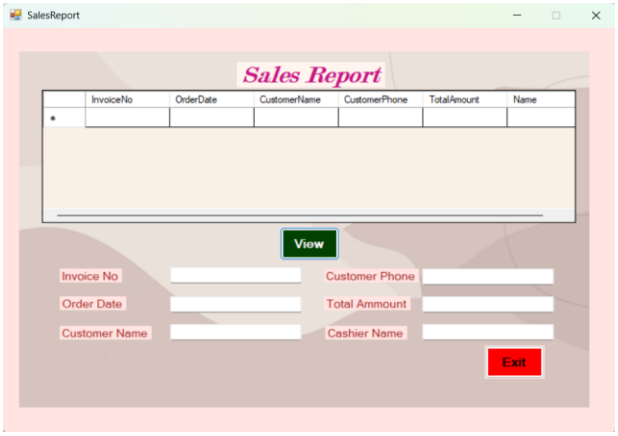


Fig-7: Sales Report View

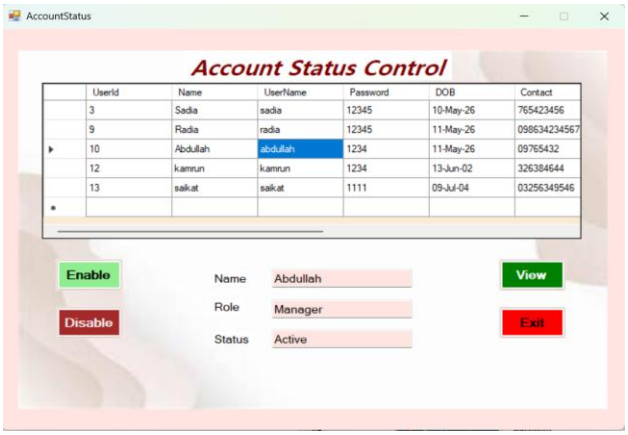


Fig-8: Admin Account Status Control for Employee

Inventory Manager Accessed Form

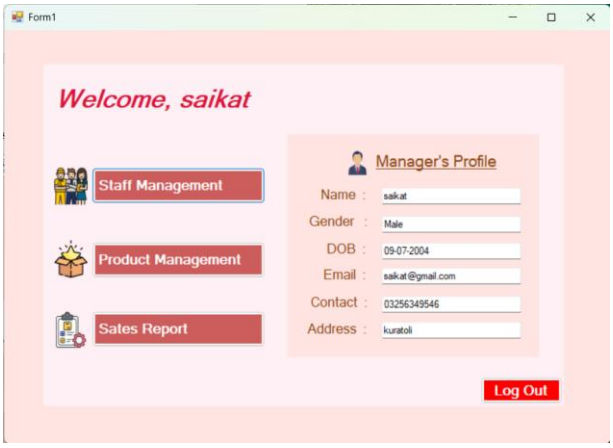


Fig-9: Manager Dashboard

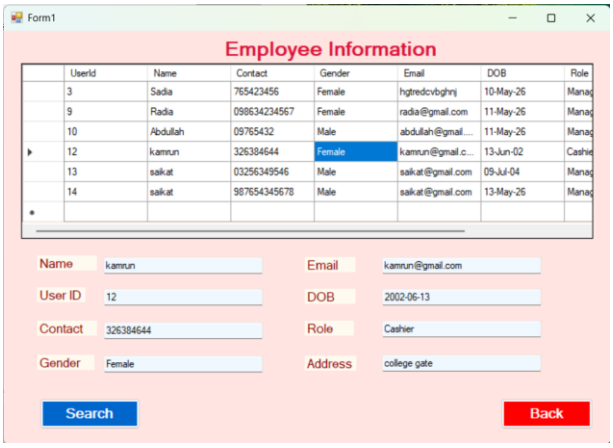


Fig-10: Staff Information View

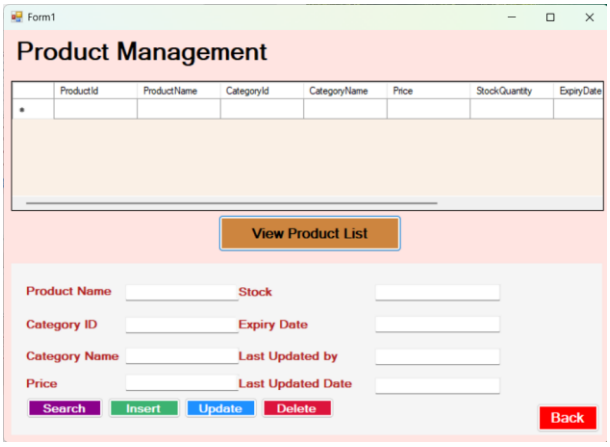


Fig-11: Product Insert, Update, Delete, View, Search

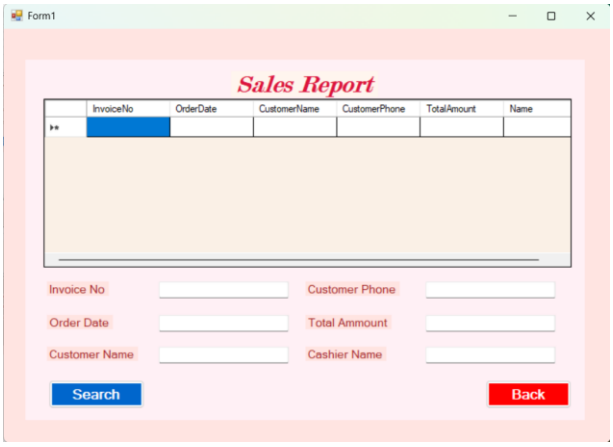


Fig-12: Manager Sales Report View

Sales Cassier Accessed Form

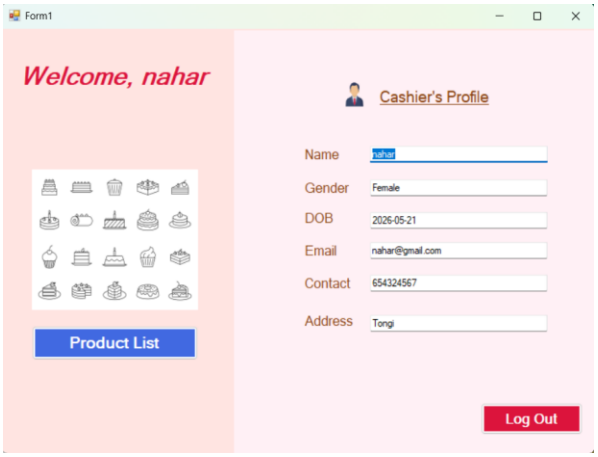


Fig-13: Cassier Dashboard

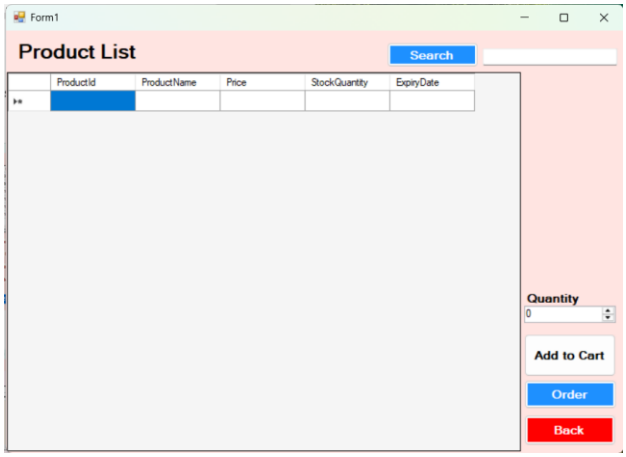


Fig-14: Product List for Order

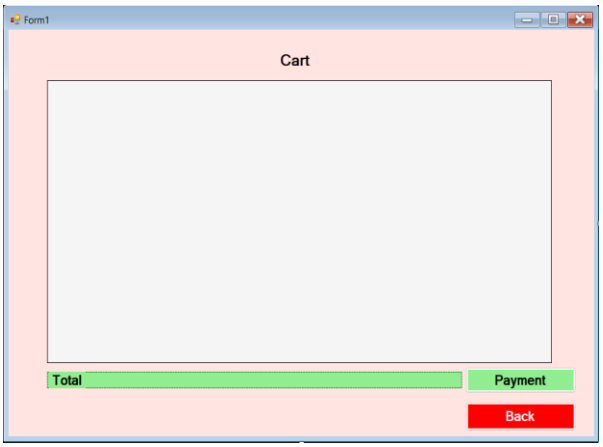


Fig-15: Cart List

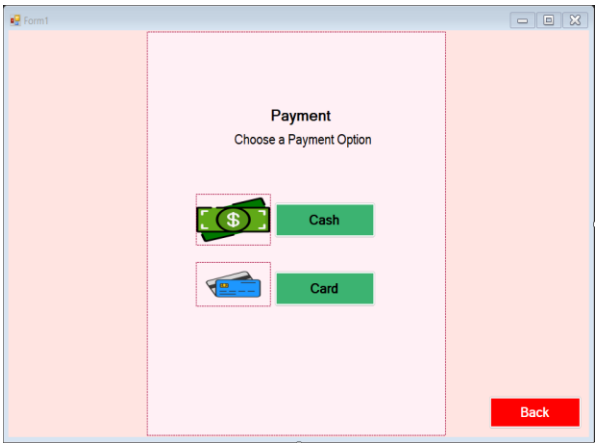


Fig-16: Payment Method Choose

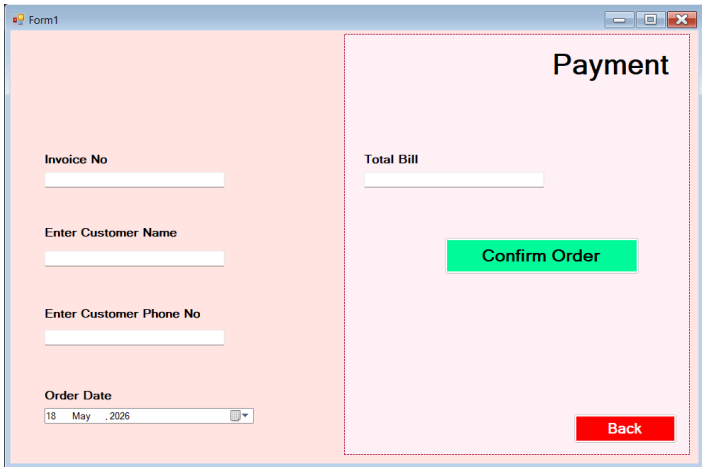


Fig-17: Payment Slip

5. Sample Code for 1 User's Insert, Update, Delete, Search

```
//INSERT
            SqlConnection conn = new SqlConnection(@"Data
Source=localhost\SQLEXPRESS;Initial Catalog=BakeryDB;Integrated Security=True;");
            conn.Open();

            string query = "INSERT INTO UserInfo (Name, UserName, Password, DOB,
Contact, Email, Role, Status, Gender, Address) VALUES('" + textName.Text + "','" +
textName.Text + "','" + textPassword.Text + "','" + dtpDOB.Value.ToString("yyyy-MM-
dd") + "','" + textContact.Text + "','" + textEmail.Text + "','" + textRole.Text +
"','Active','" + textGender.Text + "','" + textAddress.Text + "')";
            SqlCommand cmd = new SqlCommand(query, conn);
            cmd.ExecuteNonQuery();


            conn.Close();

//UPDATE
SqlConnection conn = new SqlConnection(@"Data Source=localhost\SQLEXPRESS;Initial
Catalog=BakeryDB;Integrated Security=True;");
conn.Open();

string query = "UPDATE UserInfo SET Name='" + textName.Text + "',Password='" +
textPassword.Text + "',Contact='" + textContact.Text + "',Email='" + textEmail.Text
+ "" +
    "',Role='" + textRole.Text + "',DOB='" + dtpDOB.Value.ToString("yyyy-MM-dd") +
"',Gender='" + textGender.Text + "',Address='" + textAddress.Text + "' WHERE
UserId=" + Convert.ToInt32(id);
SqlCommand cmd = new SqlCommand(query, conn);
cmd.ExecuteNonQuery();

conn.Close();


//DELETE
SqlConnection conn = new SqlConnection(@"Data Source=localhost\SQLEXPRESS;Initial
Catalog=BakeryDB;Integrated Security=True;");
 conn.Open();
 string query = "DELETE FROM UserInfo WHERE UserId=" + Convert.ToInt32(id);
 SqlCommand cmd = new SqlCommand(query, conn);
 cmd.ExecuteNonQuery();

 conn.Close();


//SEARCH
            SqlConnection conn = new SqlConnection(@"Data
Source=localhost\SQLEXPRESS;Initial Catalog=BakeryDB;Integrated Security=True;");
            conn.Open();

            string query = "SELECT * FROM UserInfo WHERE Role!='Admin' AND (Name LIKE
'%" + textName.Text + "%' AND Role LIKE '%" + textRole.Text + "%')";
            SqlCommand cmd = new SqlCommand(query, conn);
            SqlDataAdapter adp = new SqlDataAdapter(cmd);
            DataSet ds = new DataSet();
            adp.Fill(ds);
```

6. Database Table Screenshot

ASUS\SQLEXP...OrderDetails | ASUS\SQLEXP...OrderInvoice | ASUS\SQLEXP... dbo.Product | **ASUS\SQLEXP...bo.UserInfo**

UserId	Name	UserName	Password	DOB	Contact	Email	Role	Status	Gender	Address
1	Mahi	mahi	1234	2004-04-04	01772960057	mahi@gmail.co...	Admin	1	Female	Tongi
2	mahira	mahira	123	2026-05-10	0997643753738	mahira@gmail....	Admin	Active	Female	Tongi Gazipur
3	Sadia	sadia	12345	2026-05-10	765423456	hgtredcvbghnj	Manager	Active	Female	kuratoli
9	Radia	radia	12345	2026-05-11	098634234567	radia@gmail.co...	Manager	Active	Female	Tongi College G...
10	Abdullah	abdullah	1234	2026-05-11	09765432	abdullah@gmai...	Manager	Active	Male	tongi
11	sadia	sadia	1234	2005-02-18	0197343673	sadia@gmail.co...	Admin	Active	Female	Tongi
12	kamrun	kamrun	1234	2002-06-13	326384644	kamrun@gmail....	Cashier	Active	Female	college gate
13	saikat	saikat	1111	2004-07-09	03256349546	saikat@gmail.c...	Manager	Active	Male	kuratoli
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

ASUS\SQLEXP...bo.UserInfo | SQLQuery1.sq...S\ASUS (52)) | ASUS\SQLEXP..

Column Name	Data Type	Allow Nulls
UserId	int	☐
Name	nvarchar(50)	☐
UserName	nvarchar(50)	☐
Password	nvarchar(50)	☐
DOB	date	☐
Contact	nvarchar(50)	☐
Email	nvarchar(50)	☐
Role	nvarchar(50)	☐
Status	nvarchar(50)	☐
Gender	nvarchar(50)	☐
Address	nvarchar(MAX)	☐
		☐

Fig-18: UserInfo Table

Column Name	Data Type	Allow Nulls
CategoryId	int	☐
CategoryName	nvarchar(50)	☐
		☐

CategoryId	CategoryName
1	Cake
4	chocolate biscuit
5	Bread
6	Pastry
7	Cookies
8	Donuts
9	Beverages
10	Snacks
11	Chocolate
12	Sandwich
NULL	NULL

Fig-19: Category Table

Column Name	Data Type	Allow Nulls
ProductId	int	☐
ProductName	nvarchar(50)	☐
CategoryId	int	☐
Price	decimal(10, 2)	☐
StockQuantity	int	☐
ExpiryDate	date	☐
LastUpdatedBy	int	☐
LastUpdatedDate	datetime	☐

ProductId	ProductName	CategoryId	Price	StockQuantity	ExpiryDate	LastUpdatedBy	LastUpdatedD...
5	Vanilla Cake	1	900.00	20	2026-05-20	1	2026-05-11 00:...
7	Chocolate Cake	1	1050.00	18	2026-05-20	10	2026-05-11 00:...
10	Red Velvet Cake	1	1200.00	9	2026-05-20	1	2026-05-11 00:...
11	Chocolate Cake	1	500.00	9	2026-06-20	1	2026-05-17 21:...
12	Milk Bread	1	80.00	40	2026-06-10	1	2026-05-18 18:...
13	Butter Bread	1	120.00	30	2026-06-12	1	2026-05-18 18:...
14	Strawberry Pastry	2	150.00	25	2026-06-15	1	2026-05-18 18:...
15	Black Forest Pas...	2	180.00	20	2026-06-15	1	2026-05-18 18:...
16	Chocolate Cook...	3	200.00	50	2026-07-01	1	2026-05-18 18:...
17	Butter Cookies	3	180.00	45	2026-07-01	1	2026-05-18 18:...
18	Glazed Donut	4	90.00	35	2026-06-08	1	2026-05-18 18:...
19	Chocolate Donut	4	110.00	30	2026-06-08	1	2026-05-18 18:...
20	Cold Coffee	5	140.00	25	2026-06-05	1	2026-05-18 18:...
21	Mango Juice	5	100.00	40	2026-06-05	1	2026-05-18 18:...
22	Chicken Patty	6	120.00	20	2026-06-09	1	2026-05-18 18:...
23	Beef Puff	6	140.00	18	2026-06-09	1	2026-05-18 18:...
24	Dark Chocolate ...	7	350.00	15	2026-08-01	1	2026-05-18 18:...
25	Chicken Sandwi...	8	160.00	22	2026-06-07	1	2026-05-18 18:...
26	Club Sandwich	8	200.00	18	2026-06-07	1	2026-05-18 18:...

Fig-20: Product Table

Column Name	Data Type	Allow Nulls
InvoiceNo	int	☐
OrderDate	datetime	☐
CustomerName	nvarchar(100)	☐
CustomerPhone	nvarchar(50)	☐
TotalAmount	decimal(12, 2)	☐
UserId	int	☐

InvoiceNo	OrderDate	CustomerName	CustomerPhone	TotalAmount	UserId
1	2026-05-12 00:...	Saikat	01751027271	850.00	10
2	2026-05-12 00:...	Saikat	48282	1200.00	1
3	2026-05-12 00:...	Jhum	16549489	2400.00	1
4	2026-05-17 21:...	Test Customer	01700000000	500.00	8
5	2026-05-17 00:...	S2	12345	500.00	1
6	2026-05-17 00:...	s3	12345679	1200.00	1

Fig-21: OrderInvoice Table

Column Name	Data Type	Allow Nulls
DetailId	int	☐
InvoiceNo	int	☐
ProductId	int	☐
Quantity	int	☐
UnitPrice	decimal(12, 2)	☐

DetailId	InvoiceNo	ProductId	Quantity	UnitPrice
1	1	5	1	850.00
2	2	7	2	2100.00
3	5	11	1	500.00
4	6	10	1	1200.00

Fig-22: OrderDetails Table

7. Problems Faced During Project Work

During the implementation and testing phase of the project, we encountered several logical and technical challenges. Initially, configuring the local SQL Server Connection String caused runtime exceptions until the proper security credentials and database paths were correctly set. Furthermore, concatenating string variables directly into SQL queries created syntax issues whenever unexpected special characters or single quotes were entered into the textboxes. Managing smooth navigation and session handling between multiple forms—such as the Admin, Inventory Manager, and Cashier dashboards—without breaking the background application flow was also quite tricky. Additionally, writing extensive validation code was necessary to prevent data type errors, such as users entering negative values for product prices or quantities. Finally, automatically refreshing the DataGridView on the user interface right after performing an Insert, Update, or Delete operation required careful event handling to avoid restarting the entire application.

8. Future Scope

In the future, the Bakery Shop Management System can be significantly upgraded by integrating advanced features to enhance its efficiency and accessibility. A barcode scanning system can be implemented to automatically scan items at the cash counter instead of entering product details manually. To improve customer experience, automated SMS or email billing features can be added to send digital invoices instantly to customers' phones. The Admin dashboard can be enhanced with graphical analytics, such as charts and graphs, to visually monitor monthly sales trends and identify top-selling items. Additionally, adding an automated expiry date notification system will help managers keep track of perishable products before they spoil.

9. Conclusion

In conclusion, the **Bakery Shop Management System** has been successfully developed as a functional desktop application using C# Windows Forms and SQL Server database. The system effectively addresses the inefficiencies of traditional manual record-keeping by automating inventory tracking, user management, and sales workflows. With dedicated role-based dashboards for the Admin, Inventory Manager, and Sales Cashier, the application ensures proper access control and data security. The normalized database architecture successfully prevents data redundancy and keeps all store transactions accurate. This project provided us with valuable hands-on experience in database management and implementing CRUD operations to solve real-world operational problems.

10. References

[1] Visual Studio Docs, "Windows Forms Application Development with C# .NET Framework," Microsoft Corporation, [Online]. Available: https://learn.microsoft.com/en-us/dotnet/desktop/winforms/. [Accessed: May 18, 2026].

[2] W3Schools, "SQL Tutorial for Relational Database Management Systems," Refsnes Data, [Online]. Available: https://www.w3schools.com/sql/. [Accessed: May 18, 2026].

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